1. What do you mean by nanomaterials? How nanoparticles are differ from their bulk form in terms of properties?
2. Explain following properties of nanoparticles in details: SVR, Blue-shift, Superparamagnetism, Quantum confinement.
3. Explain Top Down and Bottom up approaches for the synthesis of Nanomaterials with examples.
4. What do you mean by bottom up and top down approach to synthesize nanomaterials? Write short note on ball milling method.
5. Explain Principle and working of Atomic Force Microscope (AFM) with suitable diagram.
6. Explain Principle and working of Scanning Electron Microscope (SEM) with suitable diagram.
7. What are the applications of nanomaterials?

Q1. What do you mean by nanomaterials? How nanoparticles are differ from their bulk form in terms of properties?

Ans: **Nanomaterials**: Materials having at least one dimension in the range of 1 to 100 nm are called nanomaterials.

Nanoparticles: Particles having all the three dimensions in the nanometer scale (1–100 nm).

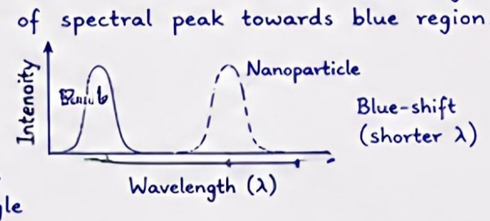
Difference between nanoparticles and bulk material:

Property	Nanoparticles	Bulk material
1. Size	1 – 100 nm	>> 100 nm
2. Surface area/volume ratio	Very high	Very low
3. Melting point	Lower than bulk	Higher
4. Electrical properties	Size dependent (may be semiconducting/conducting)	Constant
5. Optical properties	Show quantum effects, color varies with size	No quantum effects, color fixed
6. Mechanical properties	More strength and hardness	Less strength and hardness
7. Reactivity	More reactive	Less reactive
8. Magnetic properties	May show superparamagnetism	Usually ferromagnetic/ paramagnetic
9. Thermal properties	Lower thermal conductivity	Higher thermal conductivity
10. Stability	Lower stability (tendency to agglomerate)	Higher stability

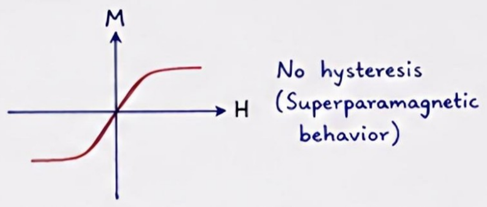
Q2. Explain following properties of nanoparticles in details: SVR, Blue-shift, Superparamagnetism, Quantum confinement.

Ans: 1. **SVR (Surface to Volume Ratio)**:
Due to very small size, nanoparticles have very large surface area per unit volume. Hence a large fraction of atoms are present on the surface which significantly affects chemical reactivity, catalytic activity and other surface dependent properties.

2. **Blue-shift**:
When the size of semiconductor nanoparticles decreases, the band gap increases. As a result, absorption/emission occurs at shorter wavelength (higher energy side) causing a shift of spectral peak towards blue region. This is called blue-shift.



3. **Superparamagnetism**:
Nanoparticles of ferromagnetic materials (Fe, Co, Ni, Fe_3O_4 etc.) behave as single domains. They show strong magnetization in presence of external magnetic field and become zero when the field is removed. They do not retain magnetization (no hysteresis).



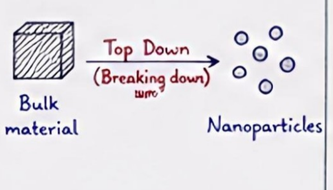
4. **Quantum confinement**:
When the particle size is comparable to or smaller than Bohr exciton radius of the material, the motion or electcers/holes is confined in all three dimensions. This leads to discrete energy levels, increase in band gap and change in optical, electrical and magnetic properties.

Q3. Explain Top Down and Bottom up approaches for the synthesis of Nanomaterials with examples.

Ans:

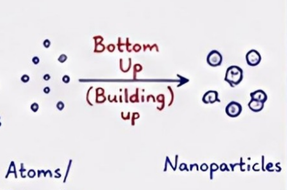
Top Down Approach:
In this approach, bulk material is broken down into nanosized particles by physical/chemical methods.

Examples:
- Ball milling
- Lithography
- Etching
- Mechanical grinding



Bottom Up Approach:
In this approach, nanoparticles are built up from atoms, molecules or clusters by chemical/physical processes.

Examples:
- Sol-gel process
- Chemical precipitation
- Hydrothermal synthesis
- CVD, Self-assembly



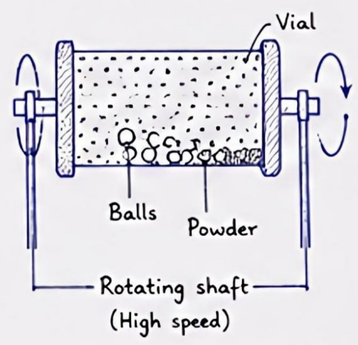
Note:
Top Down → Suitable for metals, oxides, carbon based materials.
Bottom Up → Gives better control over size, shape and purity.

Q4. What do you mean by bottom up and top down approach to synthesize nanomaterials? Write short note on ball milling method.

Ans: **Bottom up approach**:
Nanomaterials are synthesized by assembling atoms/molecules/clusters. It provides better control over size, shape and uniformity.

Top down approach:
Nanomaterials are obtained by breaking down bulk material into nanoscale particles using physical or chemical methods.

Ball milling method (Top down approach):
- In this method, bulk powder is placed in a cylindrical vial along with hard balls (steel, tungsten carbide etc.).
- The vial is rotated or vibrated at high speed.
- Due to collision and friction between balls and powder, the particles get repeatedly fractured, cold welded and re-fractured.
- This process reduces the particle size to nanometer range.
- It is simple, economical and suitable for large scale production.
- Used for metals, alloys, ceramics, oxides, carbon nanotubes etc.



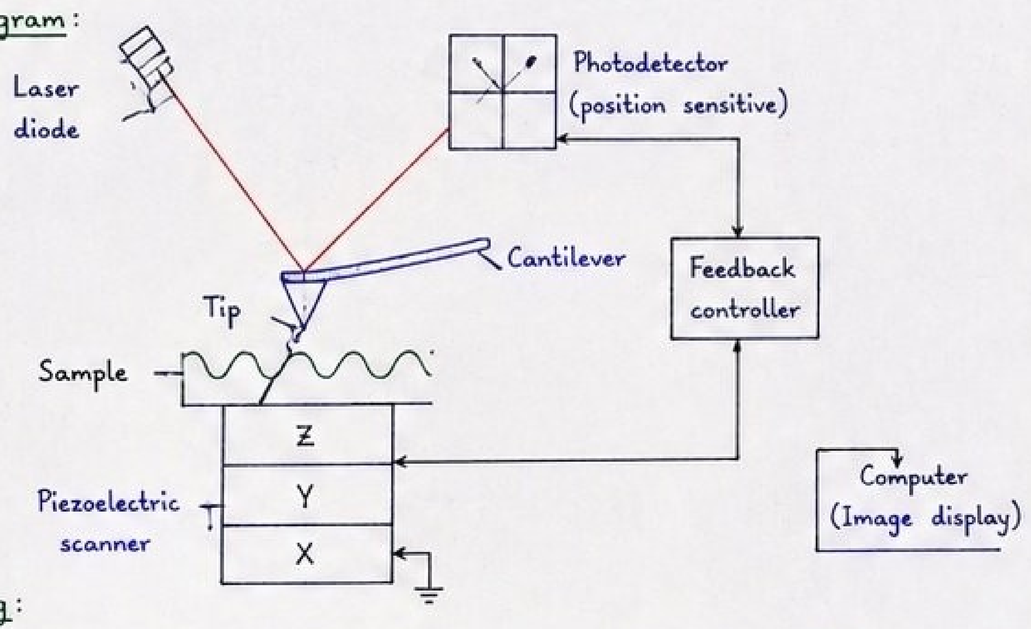
Schematic of Ball milling

Q5. Explain Principle and working of Atomic Force Microscope (AFM) with suitable diagram.

Ans: **Principle:**

AFM works on the principle of interaction between a sharp tip attached to a flexible cantilever and the sample surface.
The deflection of cantilever due to atomic level forces (van der Waals, electrostatic, etc.) is measured and used to form an image.

Diagram:



Working:

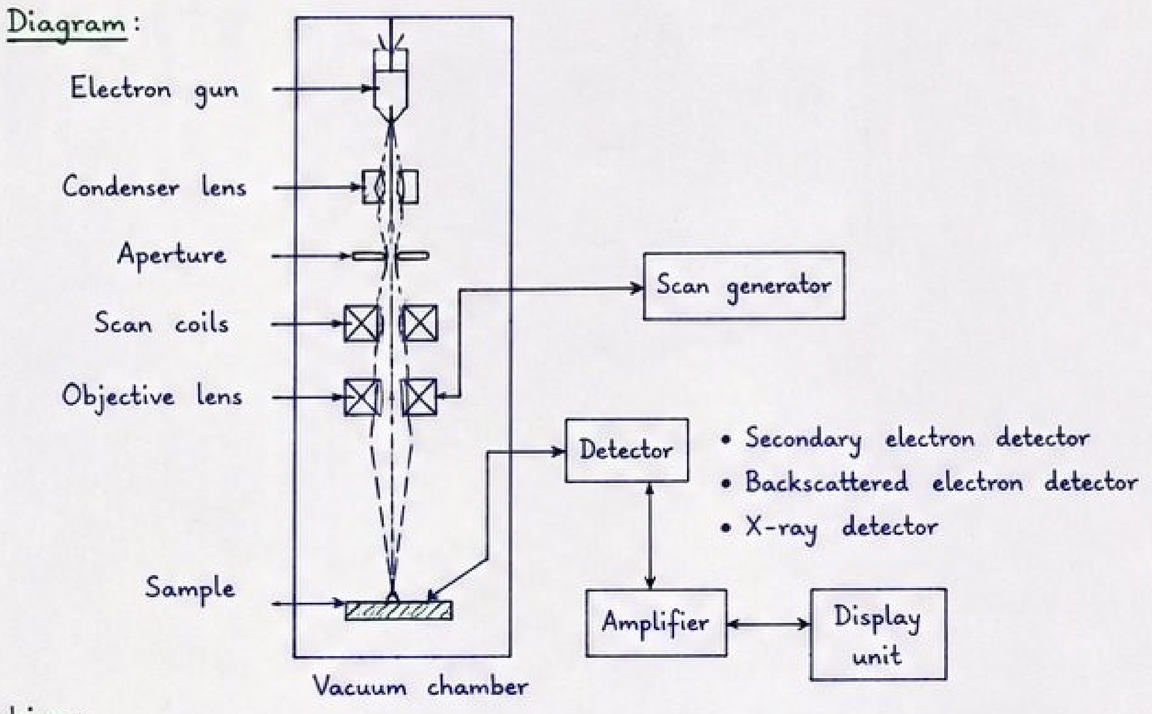
- A sharp tip at the end of cantilever scans the sample surface.
- Interatomic forces between tip and sample cause cantilever deflection.
- A laser beam reflected from the cantilever is detected by position sensitive photodetector.
- Feedback circuit adjusts the piezo scanner to maintain constant deflection.
- The vertical movement required is proportional to surface topography and gives the 3D image.

Q6. Explain Principle and working of Scanning Electron Microscope (SEM) with suitable diagram.

Ans: **Principle:**

SEM uses a focused beam of high energy electrons which scans the sample surface. The interaction of electrons with the sample produces secondary electrons, backscattered electrons and X-rays which are detected to form an image.

Diagram:



Working:

- Electron gun emits a beam of electrons accelerated by high voltage.
- The beam is focused and scanned over the sample surface.
- Electrons interact with atoms of the sample and produce secondary electrons (SE), backscattered electrons (BSE) and X-rays.
- Detectors collect these signals.
- The signal is amplified and displayed to form high resolution image.

Q7. What are the applications of nanomaterials?

Ans: Nanomaterials have wide range of applications in various fields:

1. **Electronics:** Used in nano transistors, quantum dots, memory devices, sensors and high performance batteries.
2. **Medicine and healthcare:** Drug delivery, cancer therapy, MRI contrast agents, antimicrobal coatings, tissue engineering.
3. **Energy:** Used in solar cells, fuel cells, supercapacitors, lithium ion batteries and hydrogen storage.
4. **Environment:** Water purification, air and water pollutant sensors, photocatalysis for degradation of pollutants.
5. **Catalysis:** Nanoparticles act as highly efficient catalysts in chemical reactions due to large surface area.
6. **Materials and coatings:** Used in wear resistant coatings, self cleaning surfaces, anti-corrosion coatings, nanocomposites.
7. **Cosmetics:** Used in sunscreens (TiO_2, ZnO) nanoparticles), anti-aging creams and skin care products.
8. **Textiles:** Used in stain resistant, antibacterial and UV protective fabrics.
9. **Agriculture:** Nanosensors for soil monitoring, nano-fertilizers for controlled release of nutrients.